CLASS 11-CBSE MATHEMATICS

CHAPTER: STATISTICS-SET-1

TIME : 1Hr **Mx.Marks:30**

SECTION A

Q.I Answer the following questions **[1X3 = 3 MARKS]**

1. If the mean of 3, 4, x, 7, 10 is 6, then find the value of x.
2. Find the mean deviation about the median for the data 13, 17, 16, 14, 11, 13, 10, 16, 11, 18, 12, 17.
3. The mean of a set of numbers is $\overline{x}$. If each number is multiplied by λ, then the mean of new set is.

SECTION B

Q.II Answer the following questions **[2X5 = 10 MARKS]**

4. Find the mean, variance and standard deviation for 227, 235, 255, 269, 292, 299, 312, 321, 333, 348.
5. If X and Y are two variates connected by the relation $Y = \frac{aX+b}{c}$ and Var (X) = σ^2, then write the expression for the standard deviation of Y.
6. The sum of the squares deviations for 10 observations taken from their mean 50 is 250. Find The coefficient of variation
7. Consider the numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. If 1 is added to each number, find the variance of the numbers so obtained
8. If the S.D. of a set of observations is 8 and if each observation is divided by -2, then find S.D. of the new set of observations

SECTION C

Q.III Answer the following questions **[3X3 = 9 MARKS]**

9. Find the mean deviation about the mean for the data.

Income per day	Number of persons
0-100	4
100-200	8
200-300	9
300-400	10
400-500	7
500-600	5
600-700	4
700-800	3

10. The mean of 5 observations is 4.4 and their variance is 8.24. If three of the observations are 1, 2 and 6, find the other two observations.

11. 200 candidates the mean and standard deviation was found to be 10 and 15 respectively. After that it was found that the scale 43 was misread as 34. Find the correct mean correct S.D.

SECTION D

Q.IV Answer the following questions **[4X2 = 8 MARKS]**

12. For a group of 200 candidates, the mean and standard deviations of scores were found to be 40 and 15 respectively. Later on it was discovered that the scores of 43 and 35 were misread as 34 and 53 respectively. Find the correct mean and standard deviation.

X :	35	54	52	53	56	58	52	50	51	49

13. From

Y:	108	107	105	105	106	107	104	103	104	101

the prices of shares X and Y given below: find out which is more stable in value: